

Pathology-Level ML Model Performance Summary (Test Set)
Classical ML models trained on neural network pathology predictions

Neural Network Model	Pathology	ML Algorithm	AUC Mean	AUC 95% CI	Samples	Positive Rate	Cross-Val Folds
Attention Pool Extra2	A Lines	Random Forest	0.719	[0.651-0.787]	43689	0.472	5
Attention Pool Extra2	A Lines	Logistic Regression	0.762	[0.721-0.802]	43689	0.472	5
Attention Pool Extra2	A Lines	XGBoost	0.761	[0.717-0.806]	43689	0.472	5
Attention Pool Extra2	Large Consolidation	Random Forest	0.819	[0.760-0.879]	43689	0.110	5
Attention Pool Extra2	Large Consolidation	Logistic Regression	0.899	[0.880-0.918]	43689	0.110	5
Attention Pool Extra2	Large Consolidation	XGBoost	0.866	[0.817-0.916]	43689	0.110	5
Attention Pool Extra2	Pleural Effusion	Random Forest	0.592	[0.553-0.631]	43689	0.011	5
Attention Pool Extra2	Pleural Effusion	Logistic Regression	0.771	[0.723-0.820]	43689	0.011	5
Attention Pool Extra2	Pleural Effusion	XGBoost	0.756	[0.643-0.868]	43689	0.011	5
Attention Pool Extra2	Other Pathology	Random Forest	0.601	[0.577-0.626]	43689	0.405	5
Attention Pool Extra2	Other Pathology	Logistic Regression	0.662	[0.635-0.689]	43689	0.405	5
Attention Pool Extra2	Other Pathology	XGBoost	0.650	[0.617-0.682]	43689	0.405	5
Attention Pool Extra1	A Lines	Random Forest	0.710	[0.692-0.728]	43689	0.472	5
Attention Pool Extra1	A Lines	Logistic Regression	0.751	[0.737-0.766]	43689	0.472	5
Attention Pool Extra1	A Lines	XGBoost	0.755	[0.742-0.767]	43689	0.472	5
Attention Pool Extra1	Large Consolidation	Random Forest	0.824	[0.801-0.847]	43689	0.110	5
Attention Pool Extra1	Large Consolidation	Logistic Regression	0.884	[0.871-0.897]	43689	0.110	5
Attention Pool Extra1	Large Consolidation	XGBoost	0.871	[0.850-0.892]	43689	0.110	5
Attention Pool Extra1	Pleural Effusion	Random Forest	0.655	[0.624-0.685]	43689	0.011	5
Attention Pool Extra1	Pleural Effusion	Logistic Regression	0.785	[0.738-0.832]	43689	0.011	5
Attention Pool Extra1	Pleural Effusion	XGBoost	0.815	[0.773-0.856]	43689	0.011	5
Attention Pool Extra1	Other Pathology	Random Forest	0.605	[0.592-0.617]	43689	0.405	5
Attention Pool Extra1	Other Pathology	Logistic Regression	0.655	[0.639-0.671]	43689	0.405	5
Attention Pool Extra1	Other Pathology	XGBoost	0.645	[0.640-0.650]	43689	0.405	5
Attention Pool Extra	A Lines	Random Forest	0.710	[0.683-0.737]	43689	0.472	5
Attention Pool Extra	A Lines	Logistic Regression	0.749	[0.732-0.765]	43689	0.472	5
Attention Pool Extra	A Lines	XGBoost	0.755	[0.738-0.771]	43689	0.472	5
Attention Pool Extra	Large Consolidation	Random Forest	0.819	[0.788-0.850]	43689	0.110	5
Attention Pool Extra	Large Consolidation	Logistic Regression	0.883	[0.868-0.897]	43689	0.110	5
Attention Pool Extra	Large Consolidation	XGBoost	0.867	[0.849-0.885]	43689	0.110	5
Attention Pool Extra	Pleural Effusion	Random Forest	0.664	[0.562-0.767]	43689	0.011	5
Attention Pool Extra	Pleural Effusion	Logistic Regression	0.782	[0.731-0.834]	43689	0.011	5
Attention Pool Extra	Pleural Effusion	XGBoost	0.818	[0.778-0.859]	43689	0.011	5
Attention Pool Extra	Other Pathology	Random Forest	0.608	[0.600-0.617]	43689	0.405	5
Attention Pool Extra	Other Pathology	Logistic Regression	0.654	[0.645-0.662]	43689	0.405	5
Attention Pool Extra	Other Pathology	XGBoost	0.648	[0.638-0.658]	43689	0.405	5